



***PRESENTATION FOR MINI PROJECT /
ELECTRONIC DESIGN WORKSHOP - EC681***

IoT-Based Smart Environmental Monitoring System

Real-Time Air Quality, Temperature, and Humidity Tracking

Team: Reeba Rafique · Sumyya Sadre Alam · Harshita Burnwal · Aditya Kumar

ESP32 MICROCONTROLLER

DHT11 & MQ-9 SENSORS

WI-FI DASHBOARD



TECHNO INTERNATIONAL NEW TOWN

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CERTIFICATE

This is to certify that Ms. Harshita Burnwal (18700323029) , Mr.Aditya Kumar (18700323004),Ms. Reeba Rafique (18700323041), Ms. Sumyya Sadre Alam (18700323068) has successfully completed the mini project entitled **“IOT - Based Smart Air Quality, Temperature And Humidity Monitoring System”** in partial fulfillment of the degree of Bachelor of Technology in Electronics and Communication Engineering, session (2023 – 2027) at Techno International New Town.

It is further certified that the student(s) has/ have fulfilled all the requirements laid down in Maulana Abul Kalam Azad University of Technology for the submission of the report.

Head of Department

Prof. Subhankar Bhattcharjee
Department of ECE

Supervisor Name and Signature
Prof. Md. Anoarul Islam
Department of ECE

The Problem We're Solving

Expensive

Traditional systems require costly hardware, dedicated shields, and proprietary infrastructure.

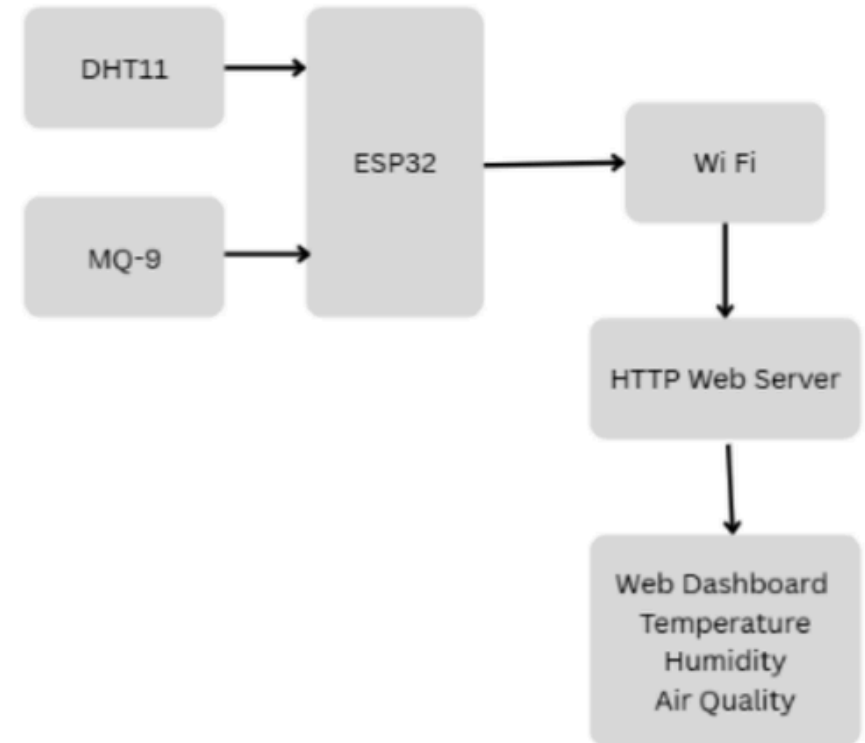
Bulky

Wall-tethered instrumentation with rigid LCD displays limits deployment flexibility.

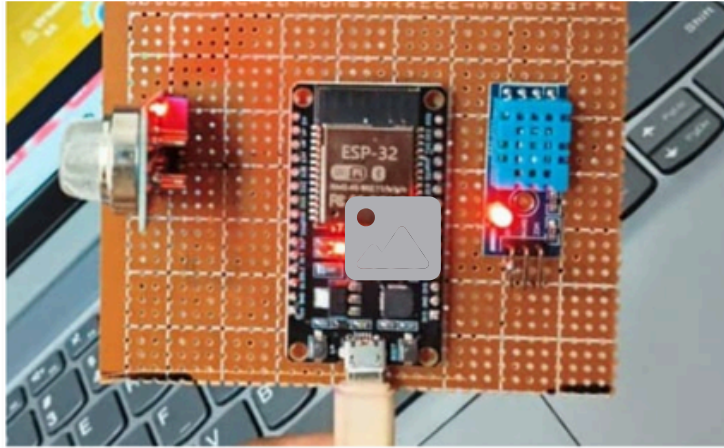
Not Accessible

Remote monitoring is difficult — data stays offline or requires complex setup.

BLOCK DIAGRAM



Our Solution at a Glance



A compact, budget-friendly prototype combining an **ESP32 microcontroller** with a **DHT11** temperature/humidity sensor and an **MQ-9** gas/air quality sensor.

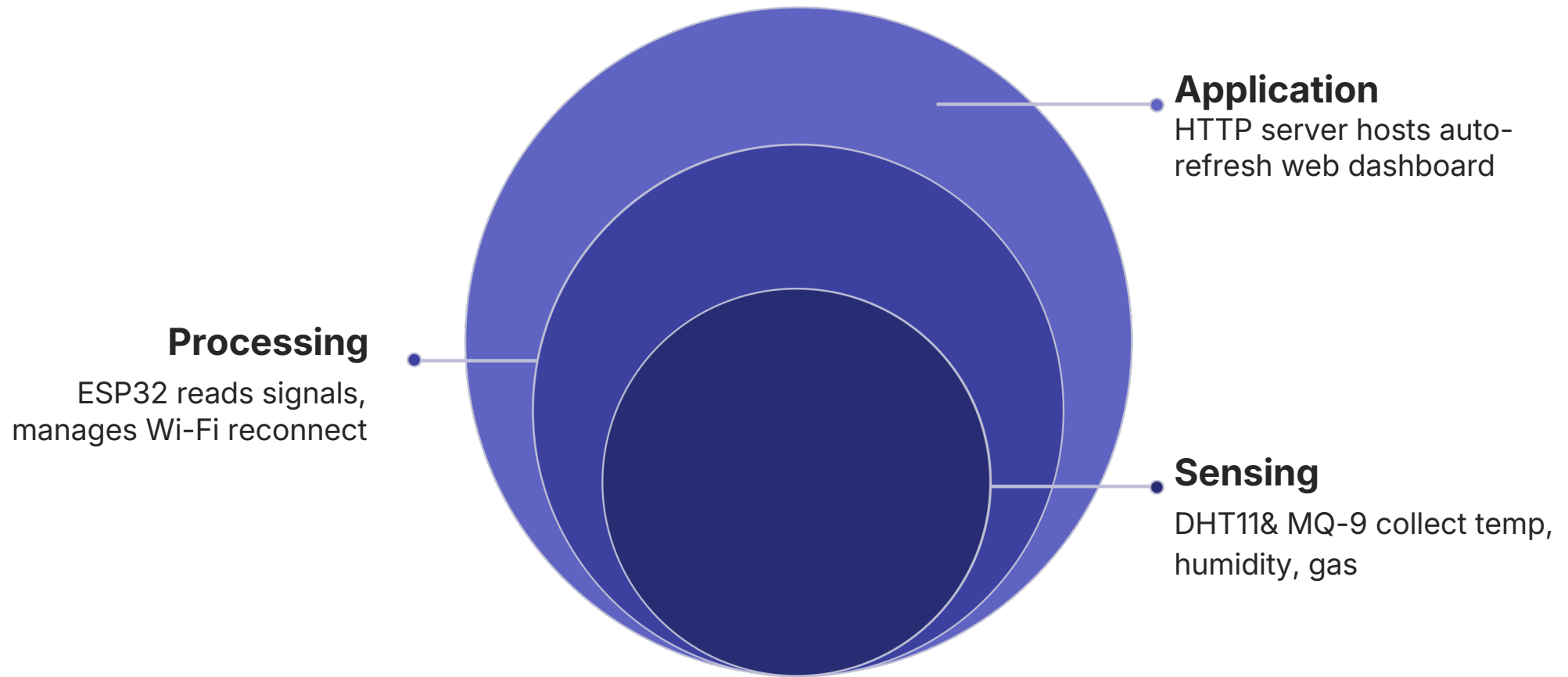
→ **Continuous Sensing** → **Local Web Server** → **Zero App Required**

Real-time ambient parameter measurement

Dashboard hosted directly on the ESP32 chip

Accessible from any Wi-Fi device via browser

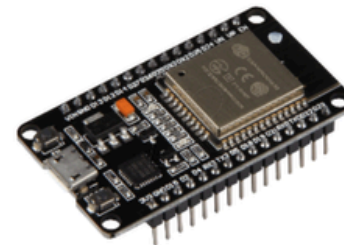
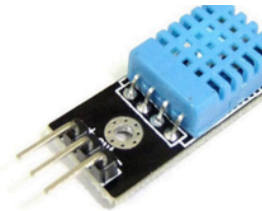
System Architecture



The sensing layer collects physical data via digital (DHT11) and analog (MQ-9) channels. The ESP32 processes signals and monitors Wi-Fi stability, triggering auto-reconnect if the link drops. The application layer hosts a lightweight HTTP server, serving live data directly to any connected browser.

Hardware Specifications & Pin Mapping

Component	Function / Parameters	ESP32 Connection
ESP32 DevKit	Dual-core CPU, integrated Wi-Fi module	Core Controller
DHT11 Sensor	Temp: 0°C–50°C · Humidity: 20%–90% RH	GPIO 4 (Digital)
MQ-9 Sensor	Detects CO, Methane (CH ₄), and LPG	GPIO 34 (Analog ADC)



Software Logic & Firmware Flow

01

Boot & Initialise

ESP32 sets up GPIO pins, connects to Wi-Fi, and launches the HTTP server on Port 80.

02

Async Sensor Polling

An asynchronous timer reads DHT11 and MQ-9 every **2 seconds** without blocking network tasks.

03

Error Handling

Sensor failures are logged to the Serial Monitor — the system continues running without crashing.

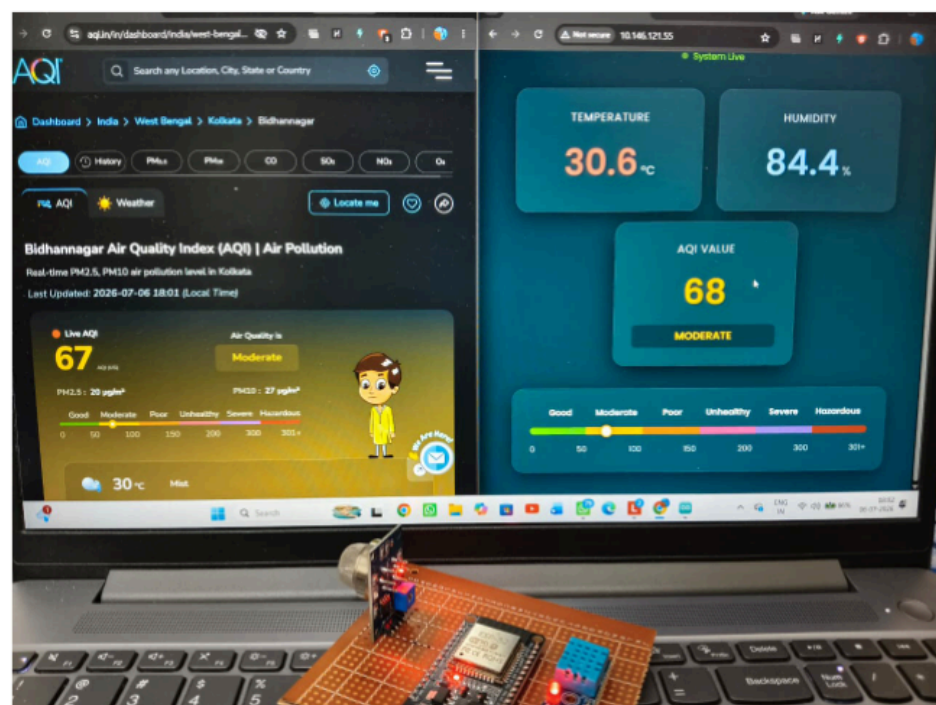
04

Dynamic Dashboard

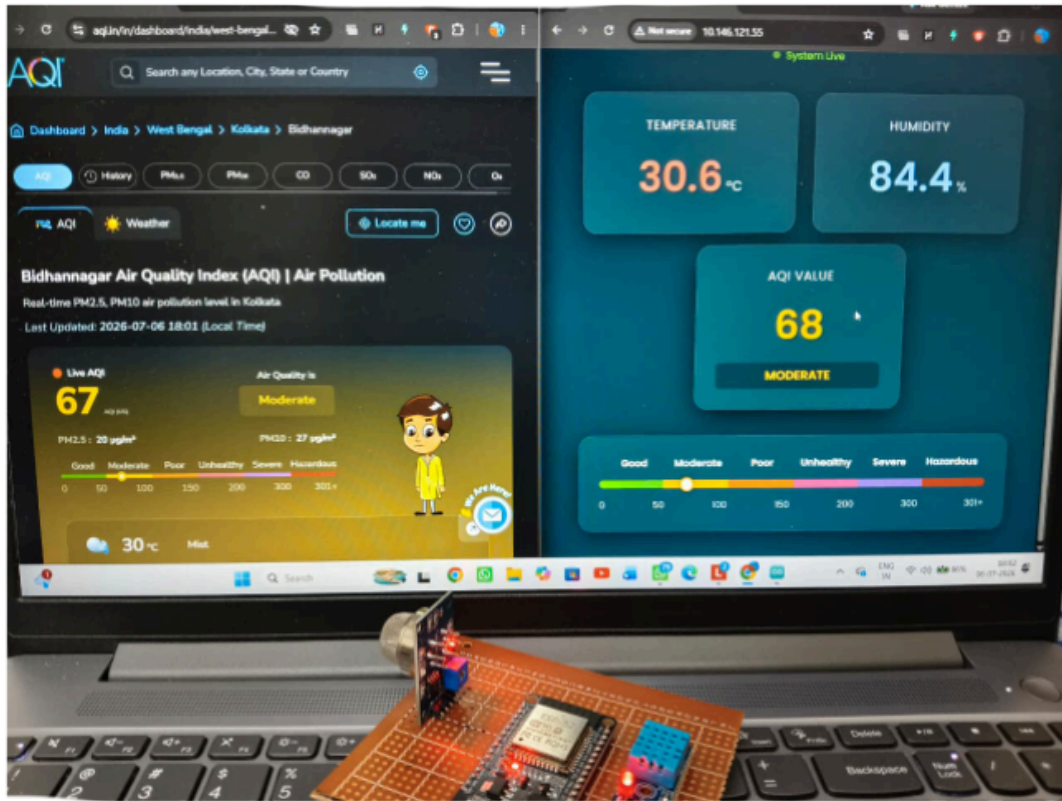
HTML/CSS/JS dashboard refreshes via AJAX — no full page reloads.

Color-

coded warnings show "MODERATE" or "HAZARDOUS" alerts.



Web Dashboard Features



Responsive UI

CleanHTML/CSSinterface adapts to phones, tablets, and desktops — no app installation needed.

AJAX Live Refresh

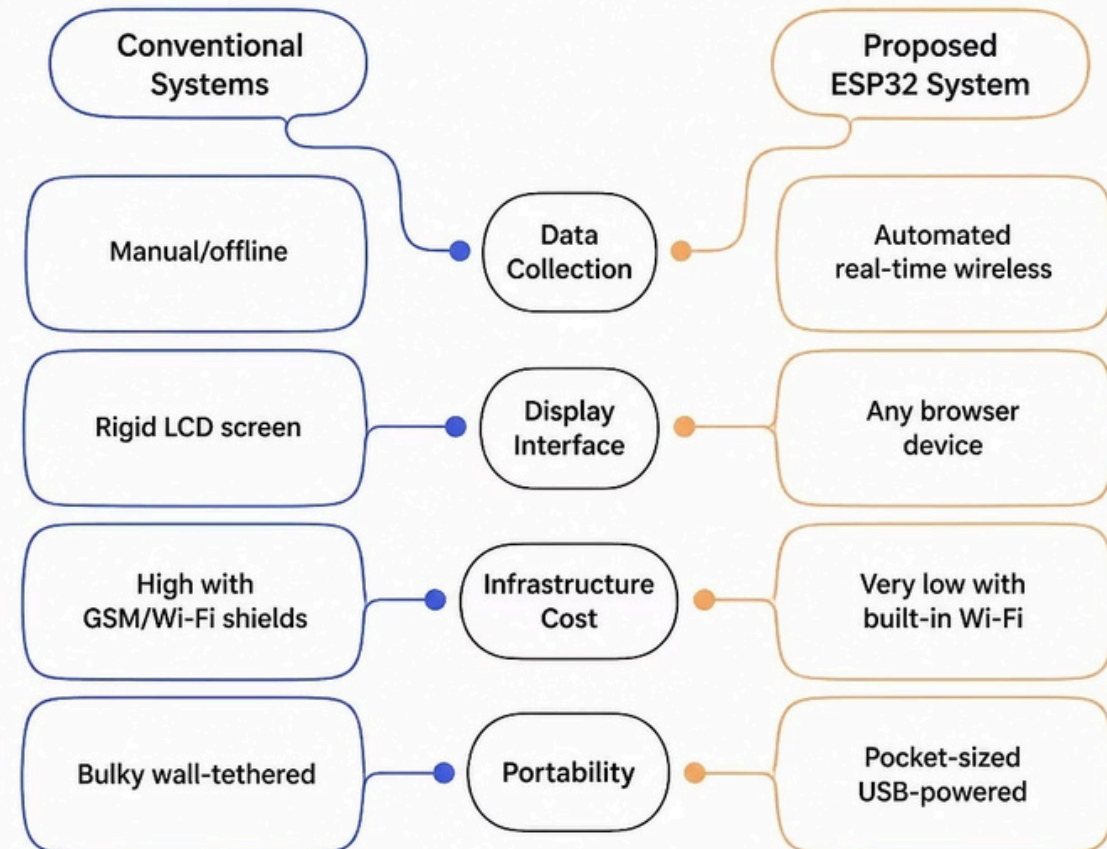
Background requests update sensor values every 2 seconds without interrupting the user experience.

Color-Coded Alerts

Dynamic warningbanners classify air quality as Safe, Moderate, or Hazardous based on gas thresholds.

Performance & Comparative Analysis

The prototype maintained a stable **2-second refresh rate** under continuous client requests — demonstrating reliable real-time performance on minimal hardware.



✓ Automated Streaming

Wireless data replaces manual readings

✓ Universal Access

Any browser-enabled device works

✓ Pocket-Sized

USB-powered, highly portable

Limitations & Future Scope

Current Limitations

DHT11 has moderate accuracy — not suitable for precision applications

Local server is restricted to the LAN — no native remote/cloud access

No onboard storage for historical data tracking or trend analysis

Roadmap Ahead

Cloud Integration

ThingSpeak, Firebase, or AWS IoT for global visualisation

Precision Sensors

Upgrade to DHT22 or BME280 for higher accuracy

Auto-Response

Relay modules to trigger exhaust fans on threshold breach

Mobile Alerts

Dedicated app with push notifications for hazardous gas spikes

Bibliography

- [1] A. Bahga & V. Madiseti, Internet of Things: A Hands-On Approach, 2015.
- [2] L. Atzori et al., "The Internet of Things: A Survey," Computer Networks, 2010.
- [3] G. Parmar et al., "An IoT Based Low Cost Air Pollution Monitoring System," IEEE RISE, 2017.
- [4] WHO, Overview of Methods to Assess Population Exposure to Ambient Air Pollution, 2023.
- [5] US EPA, Technical Reports on Air Sensor Technology, 2024.

Thank You

We appreciate your time and attention. This project demonstrates how affordable, open-source hardware can deliver meaningful real-world environmental monitoring.

Reeba Rafique

Sumyya Sadre Alam

Harshita Burnwal

Aditya Kumar



Platform: ESP32 DevKit · DHT11 · MQ-9 · Wi-Fi HTTP Server ·
HTML/CSS/JS Dashboard